

Puck Wang

Software Developer

Portfolio
GitHub
LinkedIn

WORK EXPERIENCE

Full Stack Software Developer Intern

01/2023 — 12/2023

Solace Inc

Kanata, ON

- Built and enhanced **Spring Boot microservices** integrated with **Solace event brokers**, supporting micro-frontend architectures.
- Designed **RESTful APIs** enabling **real-time broker scanning**, **configuration-as-code validation**, and graph relation visualization.
- Implemented backend-driven **user behavior analytics (Mixpanel)** to support **A/B testing** and retention analysis.
- Led **Amazon RDS MySQL migration** from version 5.7 to 8.0, coordinating rollout to minimize microservices downtime.
- Collaborated with frontend, platform, and QA teams to deliver scalable, production-grade features for the Solace Cloud Platform.

Software Developer

03/2023 — 06/2023

University of Ottawa

Ottawa, ON

- Developed a high-throughput survey platform using React and Express.js, sustaining over 1,000 TPS with p95 latency of 400 ms.
- Designed and implemented a **schedule-based email microservice** using the **outbox pattern** for reliable event handling.
- Introduced **Redis-based in-memory caching**, improving read throughput and response time by approximately 10×.
- Deployed containerized services using Docker and Nginx to support scalable research workloads.

Web Developer

05/2019 — 08/2019

University of British Columbia

Kelowna, BC

- Led development of a MERN-based catering management system for Okanagan catering groups.
- Implemented role-based access control, real-time messaging, and notification services.
- Integrated AWS S3 and AWS Lambda for scalable asset storage and serverless processing.

PROJECT

Wealth Hub Agent [GitHub](#)

- Architected a **hierarchical multi-agent system** to orchestrate end-to-end financial research, analysis, and decision agents.
- Improved research throughput by 40% using **asyncio-based parallel execution** of independent agent tasks.
- Standardized LLM tool-calling via **Model Context Protocol (MCP)**, enabling reliable integration with financial data providers.
- Built a **FastAPI** and **Server-Sent Events** pipeline to stream real-time agent status and results to a Next.js dashboard.
- Implemented a **RAG pipeline** using **LlamaIndex** and **ChromaDB** for semantic search across SEC filings and financial news.

LLM-based Youtube Summarizer [GitHub](#) | [Demo](#)

- Built a **microservice-based LLM application** for automated video summarization using **Spring Boot** and **Spring AI**.
- Developed a **Whisper-powered AWS Lambda pipeline** for caption generation on videos without existing subtitles.
- Reduced service latency from 500 ms to 100 ms by adopting gRPC, while decreasing payload size by 30%.

Blog Management System [GitHub](#) | [Website](#)

- Implemented a full-stack blog platform using Next.js, tRPC, and PostgreSQL.
- Designed role-based authentication and content management using NextAuth and GitHub OAuth.
- Introduced a RAG-based knowledge assistant using LlamaIndex and OpenAI APIs.
- Reduced image load time by 80% using AWS Lambda and cloud-based asset optimization.

EDUCATION

Master of Applied Data Science

09/2021 — 08/2023

University of Ottawa

Ottawa, ON

Bachelor of Science in Computer Science, Honours Program

09/2018 — 05/2020

University of British Columbia

Kelowna, BC

TECHNICAL SKILLS

Languages	Java, Python, TypeScript, SQL
Frontend	React, Next.js, Redux, TanStack Query
Backend	Spring Boot, FastAPI, Node.js, gRPC
Databases	MySQL, PostgreSQL, Redis, LlamaIndex, ChromaDB
Infrastructure	AWS, Docker, Nginx, Terraform, Apache Kafka, DataDog, GitHub Action